



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-27

Experiment 27. Medical Diagnosis

Aim

To implement a **medical diagnosis system using Prolog based on symptoms.**

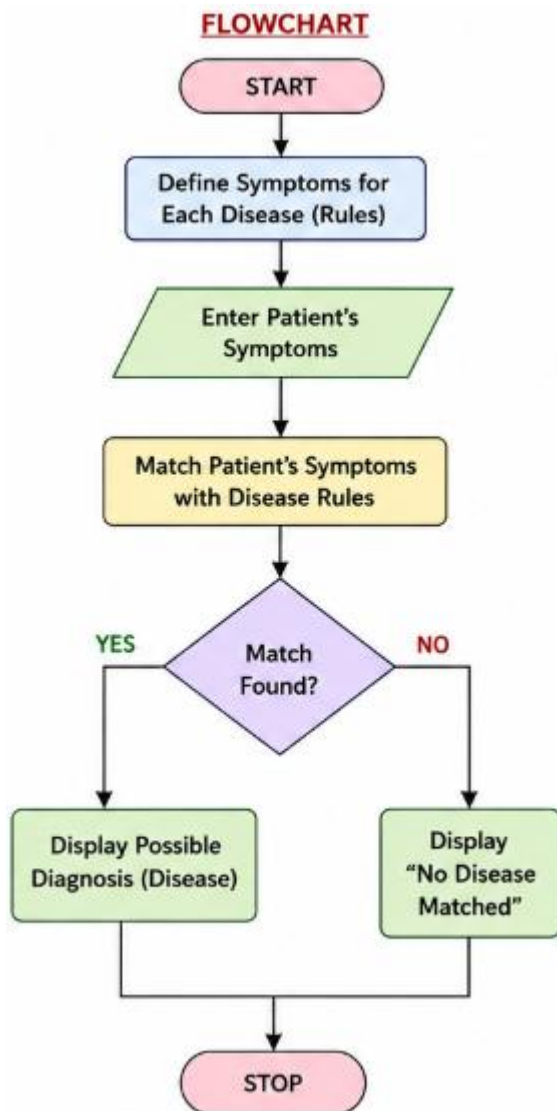
Objectives

- To understand rule-based reasoning.
- To relate symptoms with diseases.
- To identify a possible disease from given symptoms.

Algorithm

1. Define diseases and their symptoms.
2. Enter the patient's symptoms.
3. Compare the symptoms with the rules.
4. Find the matching disease.
5. Display the diagnosis.

Flowchart



Code

symptom(ravi, fever).

symptom(ravi, cough).

symptom(ravi, cold).

diagnosis(Person, flu) :-

 symptom(Person, fever),

 symptom(Person,
cough),

 symptom(Person, cold).

main :-

 Person = ravi,

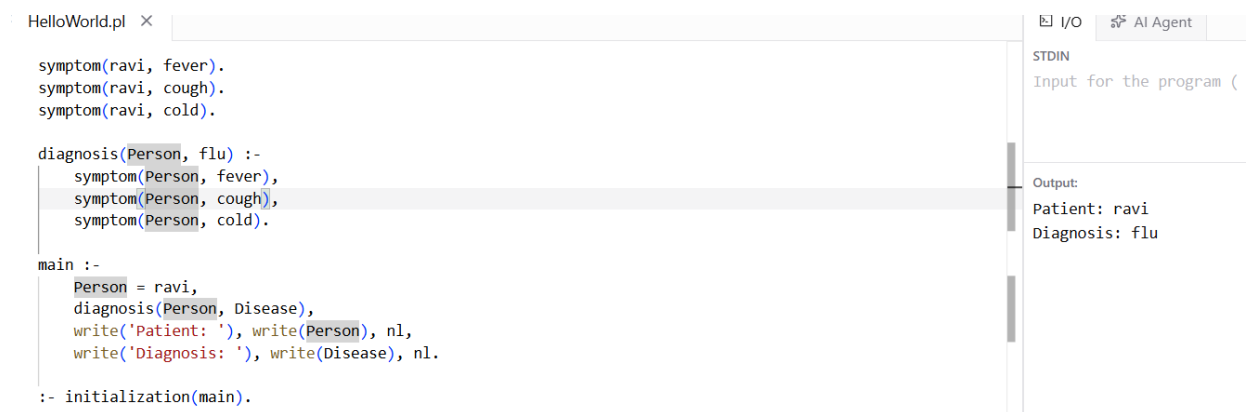
 diagnosis(Person,
Disease),

 write('Patient: '),
write(Person), nl,

 write('Diagnosis: '),
write(Disease), nl.

:- initialization(main).

Output



The screenshot shows a Prolog IDE window titled 'HelloWorld.pl'. The code defines a medical diagnosis system. The `symptom` predicate is used to record symptoms for a patient named 'ravi'. The `diagnosis` predicate is defined with a rule that checks for 'flu' based on the presence of 'fever', 'cough', and 'cold'. The `main` predicate initializes the system with 'ravi' as the patient and 'Disease' as the diagnosis, then prints the results. The output window on the right shows the results of the execution.

```
HelloWorld.pl x
```

```
symptom(ravi, fever).  
symptom(ravi, cough).  
symptom(ravi, cold).  
  
diagnosis(Person, flu) :-  
    symptom(Person, fever),  
    symptom(Person, cough),  
    symptom(Person, cold).  
  
main :-  
    Person = ravi,  
    diagnosis(Person, Disease),  
    write('Patient: '), write(Person), nl,  
    write('Diagnosis: '), write(Disease), nl.  
  
:- initialization(main).
```

I/O AI Agent

STDIN
Input for the program (

Output:
Patient: ravi
Diagnosis: flu

Result

Patient: ravi

Diagnosis: flu

Conclusion

Thus, the **medical diagnosis system** was successfully implemented using **Prolog**.